

M003 Week 7 — English Worksheet

Topic: Voxel Statue — Building Layer by Layer · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you build a statue **one layer at a time**. Each layer is a small pixel-art grid, and the layers stack **upward along y**: layer 1 at height 0, layer 2 at height 1, and so on. In the plans below, # means stone and . means air.

1 · Predict 🧠

Read each plan and set of steps. Before you imagine it happening, write what you think you will see.

```
layer 1:  # # #      layer 2:  . # .
          # # #          . # .
          # # #          . # .

build layer 1 at height 0
build layer 2 at height 1
```

Layer 1 is a full 3×3 square. Layer 2 is only the middle row. What does the statue look like from the side — wide at the bottom or wide at the top?


```
build layer 2 at height 0
build layer 1 at height 1
```

Same two layers, but the order is swapped. What does the statue look like now? Does it look like it could fall over?


```
set h to 0
repeat 4 times:
  build layer 2 at height h
  add 1 to h
```

Layer 2 (the thin middle strip) is built 4 times, each time one block higher. What shape do you get?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The statue plan says: **feet** layer at the bottom, **body** layer in the middle, **head** layer on top. Right now the statue is upside down.

```
build head layer at height 0
build body layer at height 1
build feet layer at height 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Every layer should start at the **same corner**, $x = 0$ and $z = 0$, so the statue stands straight. Right now the statue leans to one side.

```
build layer 1 at (0, 0, 0)
build layer 2 at (1, 1, 0)
build layer 3 at (0, 2, 0)
```

Hint: compare the **x** number on each line. Only the **y** number should change.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The plan has exactly **three** layers: 1, 2, 3. Right now the statue is too tall and has the wrong middle.

```
build layer 1 at height 0
build layer 2 at height 1
build layer 2 at height 2
build layer 3 at height 3
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Draw a **layer plan** for a small statue — at least 3 layers, each one a small grid — then build it layer by layer, going up one height at a time. When you finish, come back here.

Send a photo or video of your statue, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I planned my layers by ...

My statue has ... layers, and layer 1 is ...

Each layer went one higher because ...

I kept every layer at the same corner so that ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you show your statue in the world. Talk through it like you are teaching someone who has never built layer by layer. Try to use these words: **layer, height, stack, plan, grid.**

1. Show your layer plan on paper, then show the finished statue.
2. Point at the bottom layer and say its height, then point at the top layer and say its height.
3. Show how two layers are different shapes but sit on the same corner.
4. Say in your own words why the **order** of the layers matters.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.